Jonathan Jesni Manissery

AI/ML Engineer | Computer Vision & Deep Learning

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Computer Science undergraduate (Expected 2027) building practical, scalable AI pipelines. Trained a custom YOLOv8 vision model to 96.7% precision for autonomous triage and engineered a dual-encoder medical segmentation network achieving an ~0.85 F1 score.

EDUCATION

Indian Institute of Information Technology Pune, Expected May 2027 | Pune, India
Bachelor of Technology (B.Tech)
Relevant Coursework: Machine Learning, Data Structures, DBMS (SQL)

Indian School Muscat, *Higher Secondary Education (CBSE)* 05/2023 | Muscat, Oman

PROJECTS

Modified Double U-Net: Multi-Class Medical Image Segmentation,
B.Tech Project | PyTorch, Deep Learning, Medical Imaging, Computer Vision
- **Achieved ~0.85 validation F1** on severely class-imbalanced data by designing a dual-stacked Double U-Net with a combined Cross-Entropy + Multi-Class Dice loss.
- **Lifted overall segmentation F1-score by 3-4%** over single-backbone baselines by fusing VGG-19, DenseNet-121, and Xception encoders with custom Softmax attention-gating to suppress background noise.
- **Cut training time ~25%** by implementing Automatic Mixed Precision (AMP) and persistent data-loader workers across the training pipeline.

Ludex: Hybrid Game Recommendation System, *B.Tech Project | Machine Learning, Recommender Systems*
- **Improved recommendation relevance ~12–18%** over standalone models by building a hybrid engine combining collaborative and content-based filtering.
- **Increased catalog coverage ~20%** and reduced popularity bias by applying a diversity-aware re-ranking algorithm.
- **Built an ETL pipeline** over large-scale Steam user-game interaction data, ensuring 100% user-base coverage by resolving complete cold-start failures for edge-case accounts (~0.3%) via multi-tiered fallback logic.

SynthRescue: Autonomous AI Triage & Synthetic Data Engine,
Hack2skill Solution Challenge | PyTorch, YOLOv8, Computer Vision, Python, Google Cloud, Next.js, FastAPI
- **Achieved 96.7% precision** on an edge-optimized YOLOv8 model by automating bounding-box annotation through a procedural 3D data pipeline (Blender Python API).
- **Cut false positives by >56%** (raising baseline precision from 92.4%) by generating ~6,115 synthetic training images and actively applying negative-sample reinforcement on background debris.
- **Deployed a model-serving dashboard** in a 2-person team (FastAPI REST endpoint, Docker, Google Cloud), utilizing Gemini AI (LLM) to translate live drone telemetry into prioritized emergency dispatch reports in ~4.5 seconds.

SKILLS

AI / Machine Learning — Deep Learning (CNN), Computer Vision, Recommender Systems, LLMs / Generative AI / NLP, Object Detection (YOLO), Semantic Segmentation

Frameworks & Libraries — PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, FastAPI, Flask

Tools & MLOps — Docker, Google Cloud, AWS, Vercel, Git / GitHub CLI, Linux, Cisco Packet Tracer, Blender

Languages — Python, Java, C/C++, SQL

Frontend & Mobile — React.js, JavaScript, HTML/CSS, Tailwind, Dart, Flutter